

TrueNAS Administration Training

UKM Warisan Storage Infrastructure

Schedule: 9:00 AM – 4:00 PM (Lunch: 1:00 PM – 2:00 PM)

Course Objectives

1. **Understand Storage Architecture:** Master ZFS hierarchy (Disks → VDEVs → Pools → Datasets).
2. **Deploy & Configure TrueNAS:** Navigate the WebUI, configure network interfaces, and provision resilient storage pools.
3. **Manage Access & File Shares:** Configure local users/groups, enforce POSIX & NFSv4 ACL permissions, and publish secure NFS exports.
4. **Implement Data Protection:** Schedule automated snapshots, execute instant file rollbacks, and set up replication tasks.
5. **Monitor & Maintain Storage:** Interpret alerts, automate SMART & Scrub schedules, and perform a live failed-disk replacement.



Training Schedule

Time	Session / Topic
09:00 - 09:30	Course Welcome, Objectives & Lab Verification
09:30 - 10:15	Module 1: TrueNAS Architecture & ZFS Fundamentals
10:15 - 10:30	 <i>Morning Refreshment Break</i>
10:30 - 11:30	Module 2: Setup, Networking & Storage Pool Configuration (Lab 1)



Training Schedule

Time	Session / Topic
11:30 - 13:00	Module 3: Access Control & Network Sharing: NFS (Lab 2)
01:00 - 02:00	 <i>Lunch Break & Solat</i>
02:00 - 03:00	Module 4: Data Protection, Snapshots & Disaster Recovery (Lab 3)
03:00 - 03:45	Module 5: System Health, Monitoring & Disk Replacement (Lab 4)
03:45 - 04:00	Wrap-Up & Q&A

Module 1

TrueNAS Architecture & ZFS Fundamentals

(09:30 – 10:15)

What is TrueNAS?

- **Enterprise Open-Source Storage Operating System** developed by iXsystems.
- Powered by the **OpenZFS** file system (128-bit file system and volume manager).
- based on Debian operating system
- used Linux KVM virtualization and Docker container support
- Modern hardware driver compatibility and Scale-Out clustering

Why ZFS? The Foundation of TrueNAS

Traditional Storage Stack:

Hardware RAID Controller → Volume Manager (LVM) → File System (ext4/NTFS)

ZFS Unified Architecture:

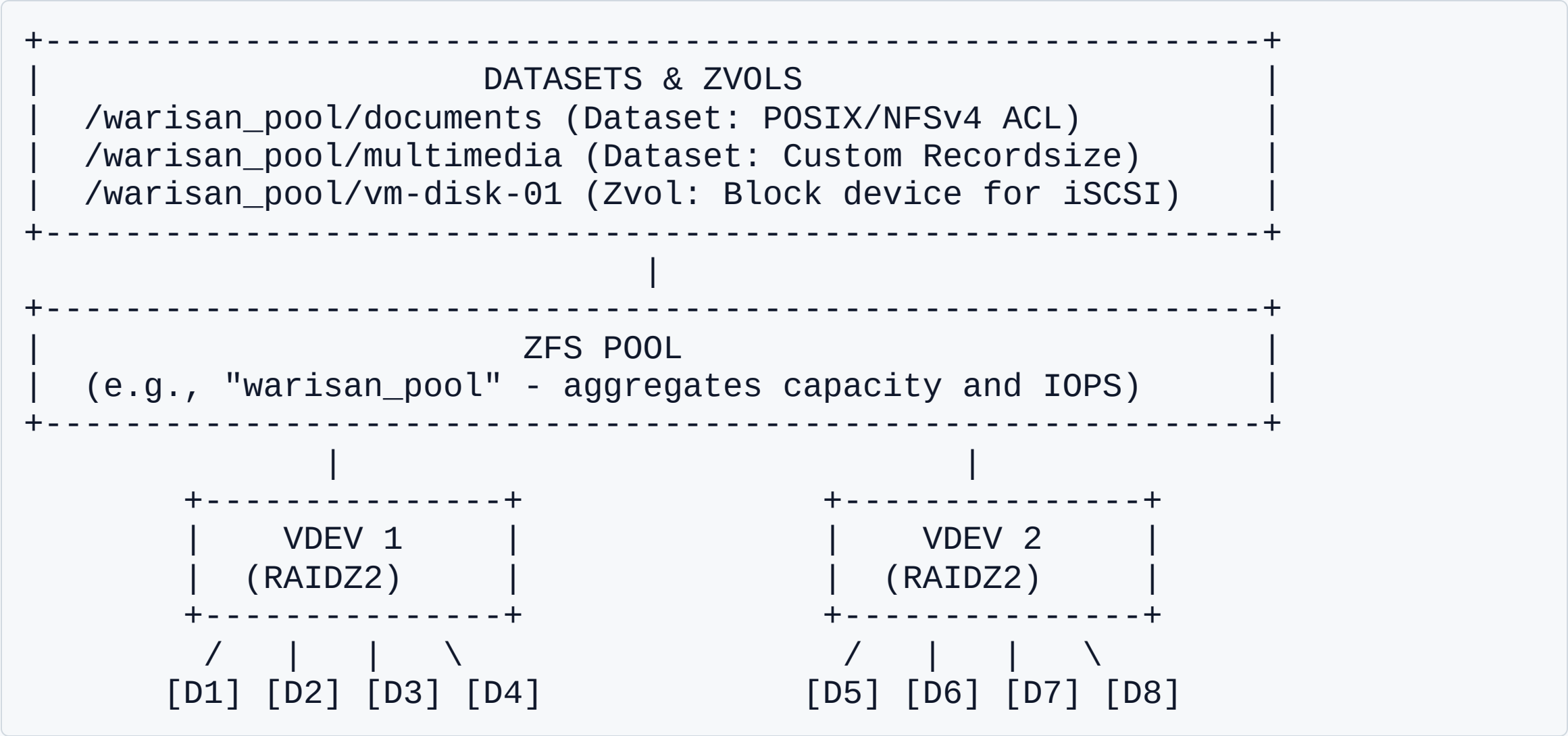
Raw Physical Disks (HBA) → **OpenZFS** (Volume Manager + File System + Cache)

Why ZFS? The Foundation of TrueNAS (cont.)

Core ZFS Superpowers:

- **Copy-on-Write (CoW):** Data is never overwritten in-place; prevents silent file corruption on crashes.
- **End-to-End Data Integrity:** Cryptographic 256-bit checksums for every data and metadata block.
- **Self-Healing:** Automatically detects silent data corruption ("bit rot") and repairs it using redundant parity.
- **Instant Snapshots & Clones:** Zero-cost, immutable point-in-time states.

The ZFS Storage Hierarchy



Understanding VDEVs (Virtual Devices)

- A **VDEV** is a logical group of physical disks providing redundancy.
- A **Pool** is composed of one or more VDEVs.
- Data is **striped** across all VDEVs in the pool.

CRITICAL RULE OF ZFS:

If a single VDEV fails with unrecoverable data loss, **THE ENTIRE POOL IS LOST!**

Never create a storage pool using non-redundant VDEVs (Stripe) for production archival data!

Redundancy Options Comparison

Topology	Min Disks	Fault Tolerance	Usable Capacity	Ideal Use Case
Mirror	2	1 disk per pair (50%)	50%	High IOPS, Databases, VMs
RAIDZ1	3	1 disk failure	$(N - 1) \times \text{Disk Size}$	Low budget, non-critical (Not recommended for $> 4\text{TB}$)
RAIDZ2	4	2 disk failures	$(N - 2) \times \text{Disk Size}$	Standard for Archives & Warisan Media
RAIDZ3	5	3 disk failures	$(N - 3) \times \text{Disk Size}$	Ultra-high capacity drives ($> 16\text{TB}$)
dRAID	≥ 8	1–3 parity + virtual spares	$(N - P - S) \times \text{Disk Size}$	Very large disk arrays; ultra-fast parallel resilvering

Datasets vs. Zvols

Datasets (File-level)

- Behaves like a rich directory or individual filesystem.
- Inherits or overrides properties: **Compression, Quotas, ACL Type, Recordsize.**
- Used for NFS and local file storage.

Zvols (Block-level)

- Presents as a raw virtual block device (`/dev/zvol/...`).
- Used as virtual hard disks for Virtual Machines or exported over **iSCSI**.

Datasets vs. Zvols (cont.)

ARC & L2ARC (Caching)

- **ARC (Adaptive Replacement Cache):** Lives in RAM. ZFS uses all free RAM as read cache!
- **SLOG (Separate ZFS Intent Log):** Dedicated ultra-fast SSD for synchronous writes.

Morning Refreshment Break

(10:15 – 10:30)

Next up: Initial WebUI Setup, Network Configuration & Storage Pool Creation.

Module 2

System Setup, Networking & Storage Pools

(10:30 – 11:30)

TrueNAS Web UI & Initial Walkthrough

- Access via browser: `https://<truenas-ip>`
- **Default Dashboard:**
 - System Information (Uptime, TrueNAS SCALE Version, CPU/Memory utilization)
 - Storage Pool Health & Capacity Gauge
 - Active Network Interfaces & Throughput
- **Key Navigation Menu Items:**
 - **Storage:** Pools, VDEVs, Disks, Disks Management, Topology
 - **Datasets:** Dataset tree, permissions, quotas, encryption
 - **Shares:** Unix (NFS), Block (iSCSI)

TrueNAS Web UI & Initial Walkthrough (cont.)

- **Credentials:** Local Users, Local Groups, Directory Services (Active Directory / LDAP)
- **Data Protection:** Periodic Snapshots, Cloud Sync, Replication Tasks, Scrub Tasks

Network Architecture & Best Practices

Key Network Concepts

- **Static IP Assignment:** Essential for servers.
- **DNS & Gateway:** Required for cloud sync, and system catalog updates.
- **Link Aggregation (LACP / bond0):** Combines multiple NICs for redundancy and bandwidth.
- **MTU (Jumbo Frames):** Standard 1500 vs 9000 MTU (Only enable if switch supports it!).

WebUI Configuration Path

1. Go to **Network** → **Interfaces**
2. Edit target adapter (e.g., `enp3s0`)
3. Uncheck **DHCP**
4. Add IP / Subnet:
`192.168.10.50/24`
5. Configure Default Route:
`192.168.10.1`
6. Add DNS: `8.8.8.8`, `1.1.1.1`
7. Click **Apply Changes & Test Changes** (60-sec auto-revert safety window).

Storage Pool Creation Best Practices

1. Keep VDEV Sizes Sane:

- RAIDZ1: 3 to 5 disks
- RAIDZ2: 6 to 10 disks (6 disks is a sweet spot for Warisan archives)

2. Never Mix Drive Capacities or RPMs inside the same VDEV:

- ZFS will limit capacity of each drive to the smallest disk in the VDEV.

3. Use CMR (Conventional Magnetic Recording) Disks:

- Avoid SMR (Shingled Magnetic Recording) drives—they can cause catastrophic pool drops during resilvering!

4. Reserve Pool Headroom:

- Keep pool utilization **below 80%** to prevent severe write fragmentation.

Dataset Properties & Optimization

Property	Default	Recommended Setting	Rationale
Compression	lz4	lz4 or zstd	Free CPU-level compression; actually <i>speeds up</i> disk I/O!
Recordsize	128K	128K (general) / 1M (video/audio) / 16K (databases)	Match chunk size to typical file sizes for archival performance.
Atime (Access Time)	On	Off	Reduces unnecessary metadata writes every time a file is read.
Case Sensitivity	Sensitive	Sensitive	Standard Linux/Unix filesystem convention.
Quota	None	Set per departmental dataset	Prevents one team from exhausting the entire repository.

Hands-on Lab 1: Pool & Dataset Setup

Objective

Create your primary storage pool `warisan_pool` and configure structured departmental datasets with compression and quotas.

Step-by-Step Tasks:

1. Navigate to **Storage** → Click **Create Pool**.
2. Name the pool: `warisan_pool`.
3. Add 4 disks into a **RAIDZ1** or **RAIDZ2** VDEV. Click **Create Pool**.

Hands-on Lab 1: Pool & Dataset Setup (cont.)

Step-by-Step Tasks:

4. Navigate to **Datasets** → Select `warisan_pool` → Click **Add Dataset**:
 - Name: `archives` | Compression: `zstd` | Recordsize: `1M` (Archival media)
5. Create a second dataset:
 - Name: `general-documents` | Compression: `lz4` | Dataset Quota: `10 GiB`
6. Verify dataset inheritance and properties in the dataset tree.

Module 3

Access Control & Network Sharing (NFS)

(11:30 – 12:45)

User & Group Management

- **Root is for Administration only!** Never use root or share credentials across users for file access.
- **Local Users & Groups:**
 - Located under **Credentials** → **Local Users & Local Groups**
 - Always assign users to specific organizational groups (e.g., `students`, `researchers`).
 - TrueNAS automatically assigns a unique UID/GID to each user/group.

```
User: 'ahmad'    —> Primary Group: 'student' —> Read/Write to /warisan_pool/archives
User: 'sarah'    —> Primary Group: 'researchers' —> Read-Only to /warisan_pool/archives
```


Permissions: POSIX vs. NFSv4 ACLs

POSIX Permissions (Classic Unix)

- Basic 3-tier model: **Owner**, **Group**, **Others**.
- Modes: Read (**r**), Write (**w**), Execute (**x**).
- Example: `chmod 770 /mnt/warisan_pool/data`
- Limitation: Cannot grant custom permissions to multiple separate groups.

NFSv4 ACLs (Enterprise Granular)

- Fine-grained Access Control Lists.
- Multiple Users and Groups can have distinct rights on the same dataset/directory.
- Supports **Inheritance** (Child files & folders automatically inherit parent rules).
- Standard access model for enterprise Unix/Linux environments.

Configuring Network File System (NFS) Exports

- **Network File System (NFS):** The primary open standard for Linux servers, HPC clusters, and archival ingestion scripts.
- **Key NFS Settings in TrueNAS:**
 - **Path:** Target dataset (e.g., `/mnt/warisan_pool/general-documents`)
 - **Networks:** Restrict access to authorized client subnets (e.g., `192.168.10.0/24`)
 - **Hosts:** Restrict access to specific client IP addresses
 - **Access Mode:** Read-Only (`ro`) vs Read-Write (`rw`)
 - **Maproot User / Maproot Group:** Security defense mapping remote client root calls to local TrueNAS user accounts (prevents unauthorized root takeover)

Mounting NFS Shares on Linux Clients

From Linux Client:

```
# 1. Install NFS client utilities
sudo apt update && sudo apt install nfs-common -y

# 2. Create mount point & mount export
sudo mkdir -p /mnt/warisan-docs
sudo mount -t nfs 192.168.10.50:/mnt/warisan_pool/general-documents /mnt/warisan-docs

# 3. Persistent mount in /etc/fstab:
# 192.168.10.50:/mnt/warisan_pool/general-documents /mnt/warisan-docs nfs defaults 0 0
```

Hands-on Lab 2: User Access Control & NFS Sharing

Objective

Create dedicated user accounts, configure ACL permissions, and export datasets securely over NFS.

Step-by-Step Tasks:

1. Go to **Credentials** → **Local Groups** → Create group `researchers`.
2. Go to **Credentials** → **Local Users** → Create user `researcher1` (set password, assign to `researchers`).

Hands-on Lab 2: User Access Control & NFS Sharing (cont.)

Step-by-Step Tasks:

3. Go to **Datasets** → Select `general-documents` → Click **Edit Permissions**:
 - Apply an **NFSv4 ACL** preset (`RESTRICTED` or `HOME`).
 - Add an ACL item: Group `researchers` → **Full Control** → Apply recursively.
4. Go to **Shares** → **Unix Shares (NFS)** → Add export for `/mnt/warisan_pool/general-documents`:
 - Set authorized network (e.g., `192.168.10.0/24` or client IP).
 - Start the **NFS Service**.
5. On your client machine, mount the NFS share and create a test directory to verify write access!

Lunch Break & Solat

(01:00 – 02:00)

Enjoy your lunch!

Afternoon session resumes at 2:00 PM with:

Data Protection, Snapshots, Disaster Recovery & Disk Maintenance.

Module 4

Data Protection, Snapshots & Disaster Recovery

(02:00 – 03:00)

The Magic of ZFS Snapshots

- A snapshot is a **read-only, frozen record** of a dataset at an exact second in time.
- **Zero Space Initially:** Takes 0 bytes when first created!
- Only consumes storage space as underlying data blocks are modified or deleted.
- Creating a snapshot takes **less than 1 millisecond**, regardless of whether the dataset is 10 GB or 100 TB.
- **Ransomware Protection:** ZFS snapshots are immutable—even if ransomware encrypts your network share, you can roll back the entire dataset in seconds!

Copy-on-Write (CoW) in Action

1. INITIAL STATE:

Snapshot "snap1" ---> [Block A] [Block B] [Block C]

2. USER MODIFIES BLOCK B TO B':

ZFS writes Block B' to a NEW physical location:

Active Dataset ---> [Block A] [Block B'] [Block C]

Snapshot "snap1" -----> [Block B] (Preserved!)

- When you delete or modify files, the old blocks remain locked by the snapshot.
- To free disk space, you must delete old snapshots that reference those blocks.

Snapshot Access via the Hidden `.zfs/snapshot` Directory

Built-in Self-Service Snapshot Recovery:

- Every ZFS dataset maintains a hidden, read-only directory: `.zfs/snapshot`
- Accessible directly over **NFS** mounts and local CLI without any special client software!

Browsing Snapshots on Client:

```
cd /mnt/warisan-docs
ls -la .zfs/snapshot
# Output:
# auto-2026-09-12_00-00
# manual-backup-01
```

Instant Self-Service Restore:

```
# Users can copy deleted or older file versions:
cp .zfs/snapshot/manual-backup-01/letter.txt ./
```

Zero downtime, zero administrator intervention!

Periodic Snapshot Tasks & Retention Policy

Configuring Automated Tasks

- Path: **Data Protection** → **Periodic Snapshot Tasks** → **Add**
- Select Dataset:
`/warisan_pool/archives`
- **Recursive**: Check to snapshot all child datasets.
- **Schedule**: Cron preset (e.g., Hourly, Daily).

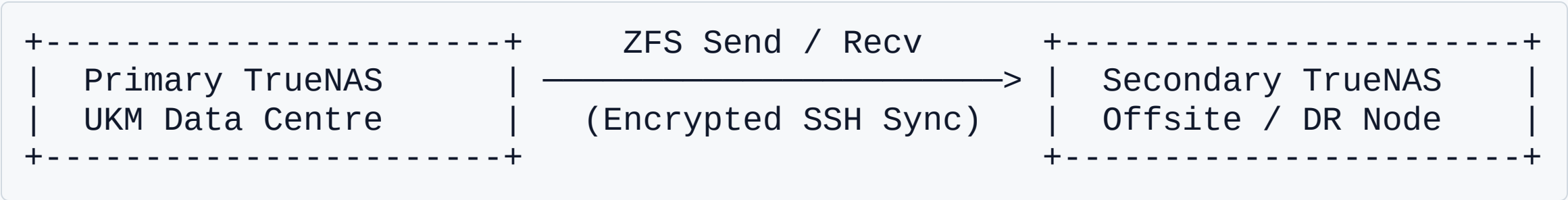
Recommended Schedule

- **Frequent**: Every 1 hour (Keep for 24 hours)
- **Daily**: Every midnight (Keep for 14 days)
- **Weekly**: Every Sunday (Keep for 8 weeks)
- **Monthly**: 1st of month (Keep for 12 months)

TrueNAS automatically purges expired snapshots according to your retention policy!

Replication Tasks & The 3-2-1 Backup Rule

3 Copies of Data | 2 Different Media | 1 Offsite Copy



ZFS Replication Advantage:

- Uses `zfs send` and `zfs recv`.
- Only transfers changed blocks (delta), not whole files.
- Extremely efficient over WAN / internet links.
- Can be pushed to an offsite TrueNAS or cloud storage via **Cloud Sync Tasks** (Backblaze B2, AWS S3, Google Cloud).



Hands-on Lab 3: Snapshots, Rollback & Recovery

Objective

Create manual and scheduled snapshots, simulate an accidental data deletion disaster, and perform an instant recovery.

Step-by-Step Tasks:

1. In your mounted NFS share (`/mnt/warisan-docs`), create a directory `important_heritage_docs` with 3 text files.
2. In TrueNAS WebUI → Go to **Datasets** → Select `general-documents` → Click **Create Snapshot**:
 - Name: `manual-pre-deletion-backup`
3. Simulate disaster: On your client, delete the entire folder (`rm -rf important_heritage_docs`)!

Hands-on Lab 3: Snapshots, Rollback & Recovery (cont.)

Step-by-Step Tasks:

4. Recover via WebUI:

- Go to **Datasets** → **Manage Snapshots**
- Find `manual-pre-deletion-backup` → Click **Rollback Dataset**.

5. Check your client mount: **All files are instantly restored!** (Also browse `.zfs/snapshot` for file-level recovery).

Module 5

System Health, Monitoring & Disk Replacement

(03:00 – 03:45)

Preventive Maintenance: Scrubbing & SMART Tests

ZFS Pool Scrubbing

- **What it does:** Reads all data blocks and verifies their 256-bit checksums.
- Automatically repairs corrupted blocks using parity.
- **Frequency:** Run **bi-weekly or monthly**.
- Path: **Data Protection** → **Scrub Tasks**.

SMART Tests

- **Self-Monitoring, Analysis, and Reporting Technology** built into hard drives.
- Detects pre-failure conditions (bad sectors, reallocated sectors, spin retries).
- **Short SMART Test:** Daily/Weekly (Tests drive electronics & heads).
- **Long SMART Test:** Monthly (Reads entire disk surface).

System Alerting & Notifications

- Never wait for a user to report a slow server!
- Configure TrueNAS to email you immediately when a drive throws errors.
- **Setup Path:**
 - i. Click the **Bell Icon** (top right) → Click **Alert Settings**.
 - ii. Set alert severity (Critical, Warning, Info).
 - iii. Go to **System Settings** → **General** → **Email**:
 - Configure SMTP Server (e.g., UKM mail gateway or institutional SMTP).
 - Send a test email to verify delivery.
 - iv. Optional: Integrate with Slack / Discord / Telegram Webhooks for instant mobile alerts.

Diagnosing Degraded Pools

```
# Check pool status via Shell / CLI
admin@truenas:~$ zpool status warisan_pool
```

```
pool: warisan_pool
state: DEGRADED
status: One or more devices could not be used because the issue is hardware-related.
action: Replace the device using 'zpool replace'.
scan: scrub repaired 0B in 01:14:22 with 0 errors on Sun Sep 01 02:14:22 2026
config:
```

NAME	STATE	READ	WRITE	CKSUM	
warisan_pool	DEGRADED	0	0	0	
raidz2-0	DEGRADED	0	0	0	
sdb	ONLINE	0	0	0	
sdc	FAULTED	8	120	0	too many errors
sdd	ONLINE	0	0	0	
sde	ONLINE	0	0	0	

Failed Disk Replacement Procedure (WebUI)

Identify Failed Disk Serial Number → Take Disk Offline → Physically Swap Drive
└→ Initiate Disk Replacement in GUI → Resilver Completes

1. **Identify the bad disk:** Note the serial number under **Storage** → **Disks**.
2. **Offline the failed drive:**
 - Go to **Storage** → Click **Manage Devices** on `warisan_pool`.
 - Select the faulted drive → Click **Offline**.
3. **Replace the hardware:** Physically pull the drive and insert the replacement disk.
4. **Initiate replacement:**
 - In TrueNAS, click on the offline disk → Click **Replace**.
 - Select the new drive from the dropdown → Confirm.
5. **Monitor Resilvering:** Watch progress bar until pool status returns to `ONLINE`.

System Configuration Backup: Golden Rule!

CRITICAL DISASTER RECOVERY TIP:

All TrueNAS system configurations (Users, Shares, Network, Pool setups) are stored in a tiny SQLite database. If your TrueNAS boot drive burns out, you can reinstall TrueNAS on a fresh drive, upload your `.tar` config backup, and be 100% restored in **5 minutes!**

How to Backup System Config:

1. Go to **System Settings** → **General**.
2. Scroll to **Save Configuration**.
3. Check **Export Password Secret Seed** (*essential for decrypting credentials!*).
4. Click **Save**: Downloads a lightweight `.tar` file to your PC.
5. Store this backup securely after every major administrative change!

Hands-on Lab 4: Disk Failure & Recovery Simulation

Objective

Simulate a hard drive failure, inspect degraded status, and execute a disk replacement and resilver.

Step-by-Step Tasks:

1. Open TrueNAS Shell or WebUI **Storage** → **Manage Devices**.
2. Simulate failure / offline:
 - Select disk `sdd` → Click **Offline** (Pool transitions to `DEGRADED`).
3. Check the Dashboard alert banner: Notice the warning notification.

Hands-on Lab 4: Disk Failure & Recovery Simulation (cont.)

Step-by-Step Tasks:

4. Run `zpool status warisan_pool` in the Shell to examine the degraded VDEV.
5. In WebUI, click **Replace** on the offline disk and select an available spare disk.
6. Observe the **Resilver** process in real-time until the pool status returns to healthy `ONLINE`.

Summary & Course Wrap-Up

(03:45 – 04:00)

Key Best Practices Checklist

- [x] **Never use Single Stripe VDEVs** for critical institutional data; use **RAIDZ2** for archives.
- [x] **Keep Pool capacity below 80%** to avoid severe performance degradation.
- [x] **Always enable Compression** (`lz4` or `zstd`) on all datasets.
- [x] **Automate Periodic Snapshots** with layered retention (Hourly, Daily, Monthly).
- [x] **Enable .zfs/snapshot visibility** so users can restore deleted files independently over NFS.
- [x] **Schedule Bi-weekly Scrubs & SMART tests** to catch silent corruption early.
- [x] **Download a Configuration Backup (`.tar`)** every time you make configuration changes.

Final Q&A

Thank You!

UKM Warisan TrueNAS Storage Administration

Keep your quick-reference guide handy!

Use the lab-preparation to simulate TrueNAS in local setting, before changing any production configuration.